

A Machine Learning model for predicting peptide properties using hierarchical graph representations

Team 9:

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Abstract

Machine learning has emerged as a powerful tool for peptide design, yet its progress remains constrained by the difficulty of predicting chemical properties for peptides composed of structurally diverse amino acids within a single, generalizable, and reliable modeling framework. As a result, most existing models are trained exclusively on linear peptides built from the 20 canonical amino acids, leaving out the non-canonical amino acids (ncAAs) that play an ever-increasing role in modern peptide chemistry and therapeutic engineering. To address this limitation, we draw inspiration from the existing hierarchical architecture and develop a combined molecular-level and residue-level framework capable of analyzing ncAAs with arbitrary chemical modifications for the identification of Antimicrobial Peptides (AMPs). Compared to sequence-based baselines, our multi-task atomic model embedded with an amino-acid-level model consistently achieves superior accuracy when classifying peptides as AMPs or non-AMPs. The resulting model demonstrates high predictive capability, thereby offering a robust computational tool for peptide engineering and drug discovery.

1 Introduction

In recent years, machine learning (ML) has emerged as a powerful tool for molecular design and discovery. Because many key molecular properties arise from the interplay of three-dimensional geometry, electronic structure, and chemical context, ML models are well-suited to learn these complex, nonlinear patterns directly from data. While various peptide models have had some successes on some levels, their approaches most commonly represent peptides solely as sequences of amino acids, without providing atom-level structural or chemical information to the model. This reliance on human-crafted descriptors to capture relationships between residues and, critically, inherently limits the number of possible amino acids to be included in the training. As a result, such models cannot naturally accommodate non-natural features, such as non-canonical amino acids (ncAAs), backbone modifications, staples, or peptide units that play an increasingly important role in modern peptide chemistry.

Incorporation of non-canonical amino acids (ncAA) has become a central strategy in peptide drug design, enabling enhanced stability, specificity, and functionality. In Fig. 1, we provide a few ncAA as references. ncAAs now play a prominent role in the peptide therapeutics market-valued at more than \$40 billion, and are key in determining diverse molecular chemical properties and biological behaviors [1]. For instance, several FDA-approved antimicrobial peptides (AMPs) incorporate D-amino acid substitutions or cycled backbones to improve protease resistance and membrane permeability. Likewise, non-natural side chains can be engineered to conjugate fluorescent labels or introduce novel chemical functionalities, supporting applications ranging from imaging to

the design of enzymes with superior catalytic performance. This expanding pool of building blocks dramatically enlarges the peptide design space; thus, it would be highly beneficial if the ML models could be applied to new amino acids with improved flexibility and generalization.

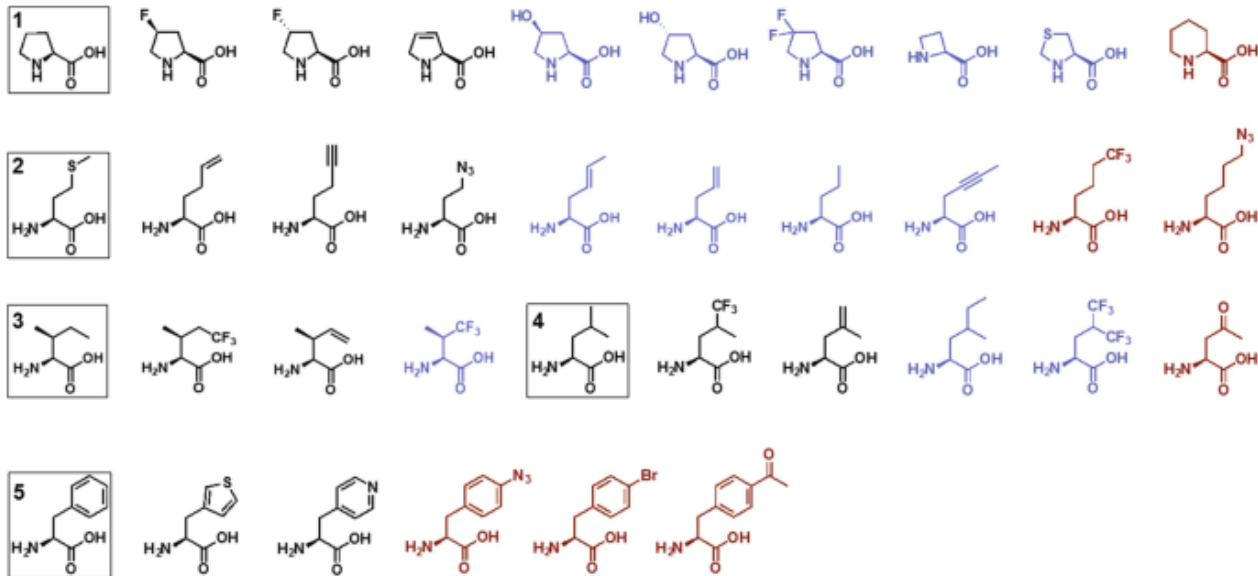


Figure 1: A few examples of modified amino acids, also known as non-canonical amino acids (ncAAs).

To address this gap, we present a redesigned atomic and residue-level layer model in the hierarchical computational framework that integrates these two levels into a unified pipeline that systematically converts modified peptide sequences, encoded through flexible SMILES-based formats, into atom-mapped RDKit molecular objects. This hierarchical model enables the extraction of fine-grained graph-based and physicochemical features, which support downstream learning tasks such as structure prediction, property estimation, and ncAA ranking. Due to time and resource constraints, we limited our project to only include the atomic and residue layers in this pipeline, with modifications and improvements to the original model.

A key innovation of this version is the implementation of non-canonical amino acids as the training data. At the atomic level, generating 3D coordinate files (XYZ format) and producing interpretable summaries of predicted favorable and unfavorable ncAAs, the updated pipeline extends the applicability of peptide ML beyond canonical residue spaces. To demonstrate its versatility, we apply the framework to datasets enriched with non-canonical building blocks and show that it consistently yields chemically meaningful feature sets suitable for a wide spectrum of machine-learning architectures. Overall, this updated implementation provides a robust foundation for generalizable peptide modeling, supporting both methodological research and practical applications in peptide design, optimization, and screening.

2 Method

For our first atomic layer GNN, we utilized PyTorch Geometric (PyG) as our main neural network library. RDKit is also applied to build molecular graphs, compute chemical descriptors, and generate atom/bond information from SMILE strings. We initially load the data sets from AAindex [2], an open source amino acid database that hosts information and chemical properties for canonical and non-canonical amino acids. Once the data is loaded, we filter out the amino acids containing

Node Feature	Description	Type	Values
Atomic Number	Identifies the element	Integer	1-118
Aromaticity	Indicates aromatic system membership	Binary	{0, 1}
Degree	Number of bonds to the atom	Integer	0-4+
Total Bonded H	Number of explicitly bonded H atoms	Integer	0-4+

Table 1: Node features description for ncAAs.

Edge Feature	Description	Type	Values
Bond Type	Chemical bond order	Continuous	1.0, 1.5, 2.0, 3.0
In Ring	Bond is part of a ring	Binary	{0, 1}
Conjugated	Bond participates in conjugation	Binary	{0, 1}
Valence Contribution (i)	Bond contribution to atom i valence	Continuous	Real values
Valence Contribution (j)	Bond contribution to atom j valence	Continuous	Real values

Table 2: Edge features description for ncAAs.

heavy elements (V, Hg, Ru), as they typically do not follow generic amino acid chemistry rules and therefore may interfere with the training process. After filtering the data, we split the data into training, validation, and test datasets as 8:1:1. The top 12 chemical properties by smallest RMSE in experimental testing are chosen as target properties for the ML prediction process. The target properties were normalized using z-score standardization: $y_{norm} = (y - \mu)/\sigma$, where μ is the mean value and σ is the standard deviation. This normalization helps to improve numerical stability during training and facilitate learning rate adjustment. Each ncAA was converted from its SMILES representation to a molecular graph using RDKit. The Chem.MolFromSmiles() function parsed the SMILES strings and created RDKit molecule objects suitable for feature extraction. The node and edge features used to represent an amino acid molecule from the atomic level are present in Table 1 and Table 2, respectively. To prevent over-conflation between different node and edge features, we checked the cross-correlation between each node and edge feature, respectively, so that we would remove the features with high cross-correlations. This helped to reduce conflation between different features, reduce the size of our model, and allowed for each feature to be trained/treated as independent.

The proposed Graph Neural Network (GNN) architecture integrates three edge-conditioned convolutional layers (NNConv) utilizing the message-passing framework [3]. Each layer operates with 64 hidden dimensions and employs a filter-generating network to modulate interactions. To introduce necessary nonlinearity for capturing complex physicochemical relationships, a LeakyReLU activation function (slope = 0.1) is applied. Regularization is enforced via a 20% dropout rate after each layer; this stochastic zeroing of neuronal activation mitigates overfitting by preventing over-reliance on specific features, thereby enhancing model generalization. NNConv uses a tiny neural network to compute the message weights based on bond features, where atoms send messages to neighbors through bonds and update embeddings accordingly. The computational flow within each block proceeds sequentially: convolution, Batch Normalization, Leaky ReLU activation, and dropout. This specific ordering is designed to stabilize training by averting exploding activations and vanishing gradients, ensuring efficient convergence. Following the three convolutional blocks, a final Multi-Layer Perceptron (MLP) processes the resulting feature matrix, outputting a sequence of length-12 vectors representing the predicted amino acid features. Training is conducted with a batch size of 96 to ensure stability. Optimization employs a ReduceLROnPlateau learning rate scheduler, which attenuates the learning rate by a factor of 0.1 upon validation loss stagnation, bounded by a mini-

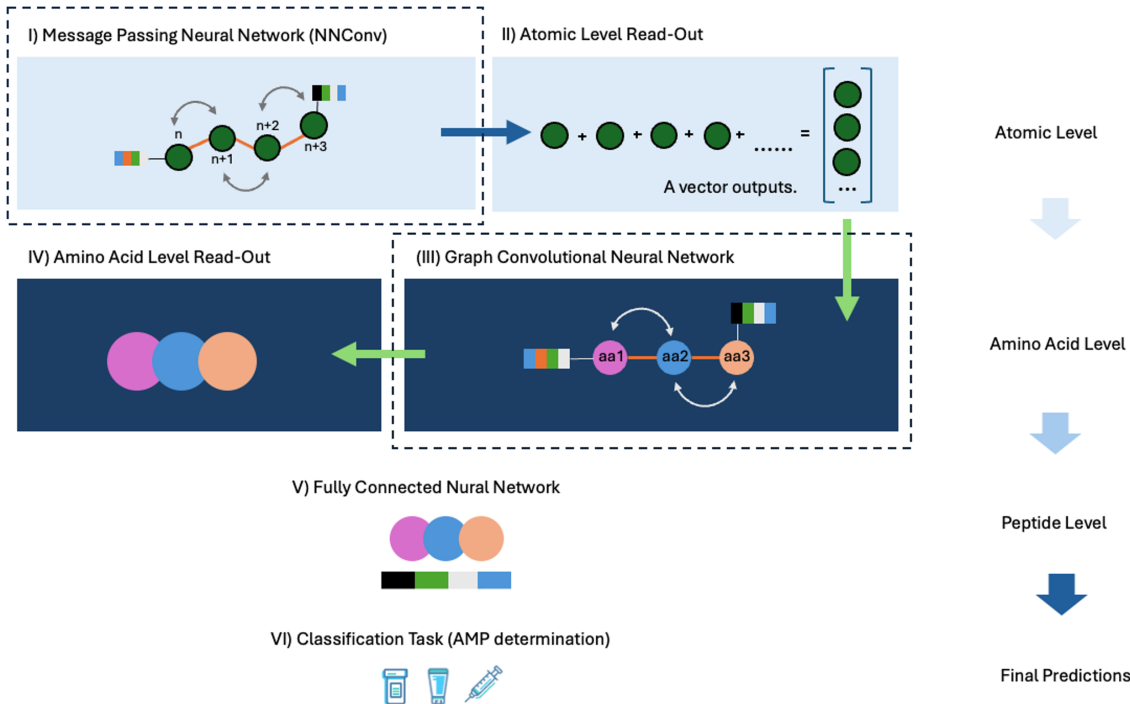


Figure 2: Diagram of model structure. The dataset is trained using a 3-layer atomic-level NNConv GNN and a 2-layer residue-level SAGEConv GNN for the AMP classification task.

num rate of 1×10^{-8} . The model is trained for a maximum of 300 epochs using a Mean Squared Error (MSE) loss function. Post-training, model efficacy is evaluated by inspecting loss and accuracy metrics relative to the multi-dimensional property vectors of the ncAAs.

To bridge the atomic-level representation with the overall hierarchical modeling objective, a residue-level peptide model was constructed using a sequence-to-graph transformation coupled with GraphSAGE-style message passing. An illustration of our model is plotted in 2. In this formulation, each peptide is represented as a graph where nodes correspond to amino acid residues and edges reflect sequential connectivity, enabling scalable learning at the peptide level without explicitly resolving atomistic detail. The dataset was partitioned into training and validation sets using an 8:2 split ratio to evaluate model generalization. The architecture integrates two stacked SAGEConv layers that utilize a message-passing framework: at each layer, a node aggregates feature vectors ("messages") from its immediate neighbors to update its own embedding, thereby capturing local sequential context. This iterative aggregation is performed with a hidden dimensionality of 64 at each layer. A dropout rate of 20% is applied after each convolutional layer to mitigate overfitting. The model outputs a single scalar in the range 0-1, corresponding to the predicted class probability. Training is performed with a batch size of 64 and a fixed learning rate of 1×10^{-3} for a maximum of 300 epochs, utilizing a logit-based loss function to optimize the classification objective.

3 Discussion

3.1 Performance of atomic-level GNN

In the atomic-level GNN framework, the selection of node and edge atomic features was validated via correlation analysis, as illustrated in the correlation matrix Fig. 3. Leveraging these features, the

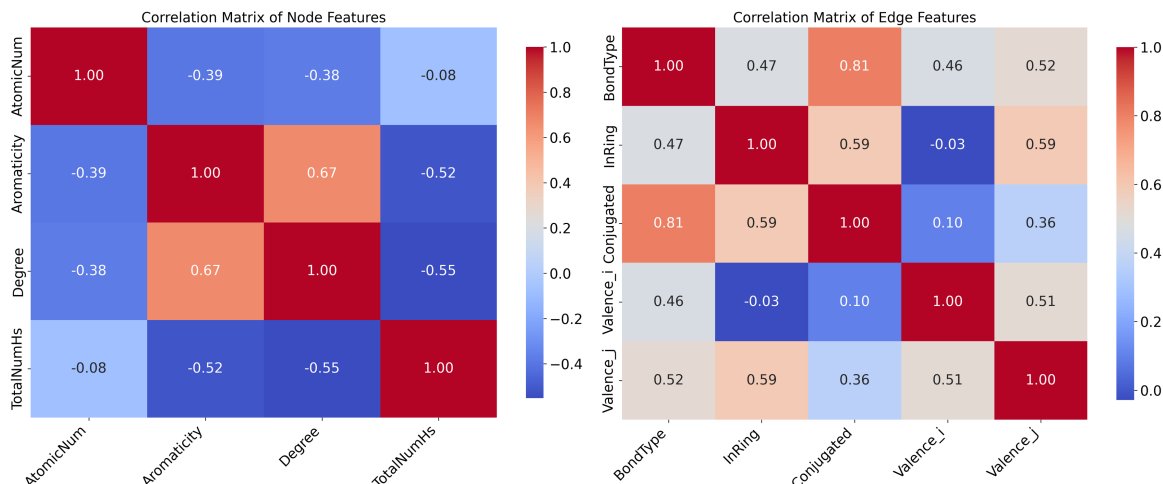


Figure 3: Correlation matrix of node and edge features for graph representation construction for atomic-level GNN.

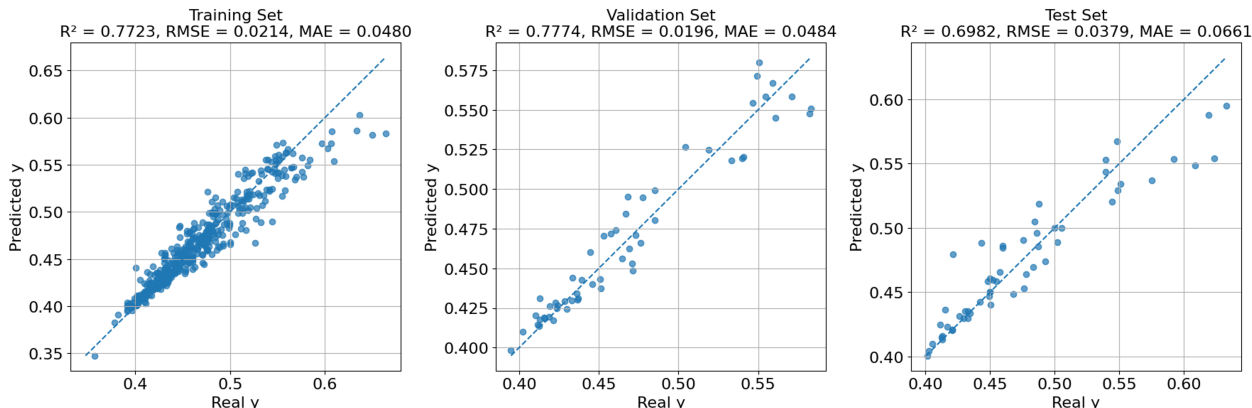


Figure 4: The agreement between predicted and actual values for the amino acid property "CHAM820101" across the training, validation, and testing subsets in a multi-objective task.

model demonstrated robust generalization capabilities across the dataset. The performance metrics for the multi-objective task are summarized as follows:

- **Training:** $R^2 = 0.7723$, RMSE = 0.0214, MAE = 0.0480
- **Validation:** $R^2 = 0.7774$, RMSE = 0.0196, MAE = 0.0484
- **Test:** $R^2 = 0.6982$, RMSE = 0.0379, MAE = 0.0661

Fig. 4 illustrates the agreement between predicted and actual values for the amino acid property "CHAM820101" across the training, validation, and testing subsets in a multi-objective task; this property specifically quantifies the polarization parameters of amino acid residues. The gap between validation and test performance suggests that a small subset of held-out ncAAs remains difficult, which is expected given the chemical diversity of modifications (e.g., rings, bulky substituents, uncommon heteroatom patterns) and the limited sample size for rare motifs. Importantly, because NNConv conditions messages on bond features, it is naturally suited to capture local

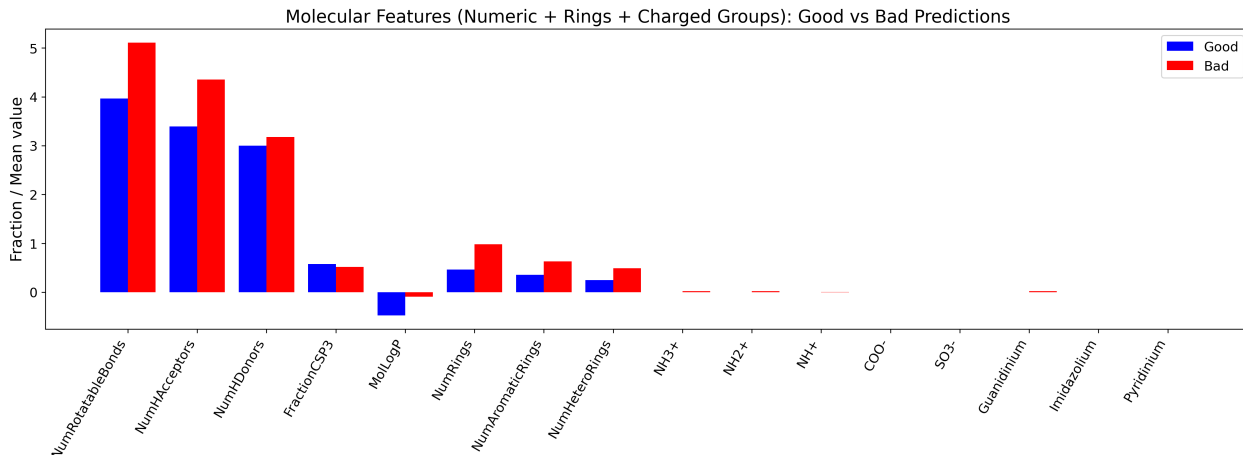


Figure 5: Amino acid features for good and bad predictions at the atomic-level GNN.

electronic/structural context that sequence-only baselines cannot represent (e.g., conjugation, ring membership, and bond order effects). This advantage becomes most visible on heavily modified residues where the same "amino-acid token" can correspond to very different chemistry under ncAA extensions. As a sanity check, we also trained an atomic model on a single representative property. In this simpler setting, performance is significantly higher, reaching **test** $R^2 \approx 0.909$ (with similarly strong train/validation scores), demonstrating that the architecture can fit structure-property relationships reliably when the target is not multi-objective.

For error analysis, we rank ncAAs by absolute prediction error and export representative structures into XYZ files for visual inspection. Empirically, we find that *most of the test error is concentrated in a small number of ncAAs*, rather than being evenly distributed. To further probe this, we compared global RDKit descriptors (molecular weight, rotatable bonds, H-bond donors/acceptors, ring counts, etc.) between well- and poorly-predicted groups. Interestingly, these coarse descriptors show broadly similar ranges across the two groups, suggesting that the hard cases are not explained by simple size or polarity alone. The distribution of accurate versus inaccurate predictions across various feature categories is detailed in Fig. 5. A plausible interpretation is that the difficult ncAAs rely on *subtle local chemical context* (e.g., specific functional group placement, conjugation patterns, and rare substructures) that are underrepresented in the training distribution. For instance, several misclassified instances contain charged functional groups whose specific electrostatic contributions may not be fully captured by the current graph representation. This motivates two concrete future improvements: (i) adding data augmentation via multiple conformers and (ii) adopting geometry-aware (equivariant) GNNs when reliable 3D structures are available.

3.2 Performance of residue-level GNN

In our implementation, the residue-level model reaches strong discrimination performance (validation AUC approaching ~ 0.93 in one training run, and test AUC ~ 0.95 in a lightweight experiment), showing that peptide-level prediction can be accurate even with simplified inputs. In Fig. 6, we provide the confusion matrix in our validation dataset. The matrix reveals a balanced classification profile, though the slightly higher rate of false positives suggests the model is somewhat conservative in predicting antimicrobial activity. However, the residue-only representation cannot naturally generalize to arbitrary ncAA chemistry without additional mechanisms (e.g., mapping new residues to learned atomic embeddings). This contrast supports the central motivation of our project: *atom-*

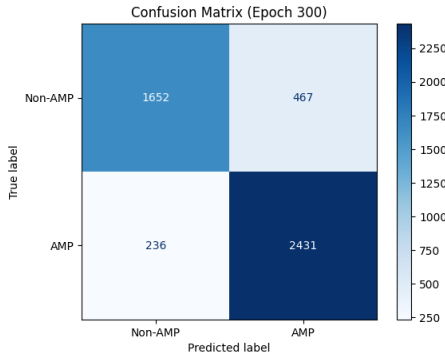


Figure 6: Confusion matrix in the classification of AMPs from our constructed hierarchical graph representation.

level modeling is essential for generalization across chemically diverse ncAAs, while residue-level models remain useful for peptide-scale tasks once suitable residue embeddings are provided.

4 Conclusion

We developed and tested a combined atomic-level and residue-level framework aimed at extending peptide machine learning beyond the 20 canonical amino acids to include non-canonical amino acids with arbitrary chemical modifications. Our main contribution is an **atomic multi-task GNN** that consumes RDKit molecular graphs constructed from flexible SMILES-based ncAA representations and predicts a vector of physicochemical properties. On held-out ncAAs, the multi-task atomic model achieves **test** $R^2 \approx 0.70$ with reasonable RMSE/MAE, and a single-task ablation reaches **test** $R^2 \approx 0.91$, confirming that the architecture can capture structure–property relations strongly when the target complexity is reduced. We further performed qualitative error analysis by identifying best/worst predicted ncAAs and exporting representative structures for inspection, finding that errors are concentrated in a small set of chemically distinctive outliers.

In addition, we implemented a residue-level peptide model as a complementary component of the hierarchical pipeline. While this sequence-derived representation can perform well on peptide-scale prediction tasks, it lacks the chemical resolution needed to robustly generalize across novel ncAA modifications. Together, these results support a clear message: atom-level modeling is a key enabler for ncAA generalization, and residue-level modeling becomes most powerful when augmented with atomic-derived embeddings that carry chemically grounded information.

Future work will focus on (i) enlarging and balancing ncAA coverage to reduce out-of-distribution failures, (ii) incorporating conformer ensembles and/or geometry-aware equivariant GNNs to better capture stereochemistry and 3D effects, (iii) adding uncertainty estimation to flag unreliable predictions for novel chemistries, and (iv) completing the full hierarchical pipeline by integrating atomic-derived residue embeddings into peptide-level predictors for end-to-end peptide design and screening.

References

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